Assessment 3: Reflective Portfolio

IFN735 – Industry Project

Student: Tanvir Singh - N12285145

Project Team: 38 - MediBridge: An AI Healthcare Navigator

Role: Team Lead

Tutor: Lakmali Herath Jayarathna

Industry Mentors: Graham Fellows, Wesley Huffstutter

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Reflection 1

Mid-Stage: Strategic Technical Leadership

[Describe]

The most consequential leadership act in the first half of this unit has not been a line of code; it has been a deliberate contraction of scope. As Team Leader of Project 38, over the past several weeks I had been filtering our brainstormed problem statements down to **MediBridge - an AI Health Navigator for international students** - but the version we initially carried into our scoping session with our industry mentors, Wesley Huffstutter, was still sprawling: real-time GP wait-time prediction, five-language support, anonymous mental-health triage, Pinecone-hosted vectors, and a multi-cloud deployment plan. In that single conversation, mentor feedback combined with my own escalating awareness of complexity-versus-timeline forced a deliberate revision. We did not amputate features; we *refined and consolidated* the ambition into three defensible pillars - **an Agentic-RAG OSHC chatbot, a Smart Clinic Search with billing-type filters, and a Gap Cost Calculator grounded in Medicare Benefits Schedule (MBS) rebate logic** - bounded to **Australian-based international students and a defined subset of OSHC insurers.** That afternoon I, along with my teammates, rewrote the scope section of the Project Establishment Report around the contraction, and brought the narrower vision back to the team the same week. Sitting at the end of Week 8, with schemas drafted and knowledge ingestion underway, this decision feels like the most load-bearing thing I have done so far.

[Interpret]

I had walked into that mentor session believing that breadth signalled ambition. Looking back from where I sit now, I think breadth had actually been signalling *risk*. The cognitive dissonance at the time was uncomfortable: **my instinct as a first-time Team Lead was to protect the team's creative range, but the mentors were pointing at a different obligation - to ship something verifiable rather than something impressive.** The narrower vision, paradoxically, has *dissipated the vagueness* that had been slowing delegation. Once "what counts as done" was specifiable in three crisp sentences, the knowledge-base ingestion plan crystallised around concrete documents (the MBS XML, the OSHC Deed PDF, six insurer fact sheets), arguments about feature ownership dissolved, and teammates could begin their schema designs and research with a shared mental model. The reframe I am carrying into the build phase that begins next week is that **ambition in industry IT is not measured by features-listed; it is measured by features-shippable-with-clinical-safety.**

[Evaluate]

Read through Tuckman's lens, the scope contraction is what has compressed our *forming* into *norming* in under two weeks. The brief storming phase - teammates negotiating which research strands survived consolidation - would, in a slower project, have spilled into Sprint 3. Pairing the decision with **Scrum's MoSCoW prioritisation gave the team an explicit Must-Have backlog**, which has since been codified in the seven-sprint plan of our Project Establishment Report. The strength of the move, as I see it now, is that it has neutralised Scope Creep - a Medium-likelihood, Medium-impact risk in our register - at the earliest possible point and built mentor trust by demonstrating that the team responds to feedback architecturally rather than cosmetically. The honest limitation, even at this mid-stage moment, is that I have no leading indicator of progress beyond "the team is busy and the mentors are pleased." **If I were to grade my performance to date, I would call it HD on my leadership decisiveness and understanding of Software.** Albeit, I do not yet have story-points-completed, a burndown, or a quantitative Definition of Done for AI accuracy.

[Plan]

Three commitments follow from this insight as the build phase begins. First, before the next mentor meeting I will introduce story-point velocity tracking and a written, AI-accuracy-aware Definition of Done, so that the rigour our Project Establishment Report promises on paper actually lives in our task tracker. Second, the next time I have to make a scope call I will treat an early, evidence-based "no" as a deliverable in its own right, rather than something I drift toward only when ambition cracks. Third, I will deepen my reading on the leadership side of agile - specifically Marty Cagan's product-discovery framing and the 2020 *Scrum Guide*'s revised **distinction between *commitment* and *forecast*** - so that my future scope decisions are anchored in a richer mental model than the binary I was operating with in Week 4. The longer-term lesson I want to carry into the second half of the semester, and beyond this unit, is that **strategic leadership in IT is less about the features one chooses to build and more about the ones one chooses, deliberately and in writing, *not* to.**

Reflection 2

Late-Stage: Technical Execution and Systems Integration

[Describe]

Over the past few weeks my role on Project 38 has inverted. With the knowledge base assembled and the schemas drafted, I have shifted from delegating Scrum-Master to head-down **full-stack developer of the AgenticRAG feature *and* integration owner for the entire MediBridge web application.** In one intense fortnight I **authored the SQLite schema** (mbs_items, insurer_tiers, oshc_coverage_rules, clinics) and its FTS5 full-text indexes, ingested embeddings into a ChromaDB vector store, and **built the LangGraph ReAct loop with six bound tools** - search_mbs_items (a 60/40 vector–FTS5 reciprocal-rank-fusion hybrid), lookup_mbs_item, calculate_oshc_coverage, check_waiting_period, query_oshc_rules, and search_clinics. I presented the agent to our mentor through a CLI demo last week. Their feedback was crisp: ***define metrics, evaluate the agent, integrate the remaining features and frontend, and restructure the agent's output*** - both the mentor and the tutor flagged that the responses, while accurate, were **text-heavy, unstructured, and difficult for a user to scan or track specific information from.** I am currently designing a LangSmith evaluation harness - hallucination rate, tool-selection accuracy, latency, faithfulness, and answer relevance - still in progress as I write this. In parallel, once the UI was delivered, I integrated the team's work into a FastAPI-mounted Progressive Web App. The critical incident during that integration came only a few days ago: **a logical bug in the Gap Cost Calculator, where the out-of-pocket formula was mishandling NULL and exact-charge edge cases.** I caught it during demo preparation for this week, not in the original unit tests.

[Interpret]

Donald Schön's distinction between *reflection-in-action* and *reflection-on-action* maps neatly onto these recent artefacts. **The Gap Cost NULL bug was reflection-in-action** - a demo-eve realisation that the formula's branching collapsed when charged fee was absent. I rewrote the precedence logic in a single sitting because the leader-turned-builder vantage point now reveals the full XML-through-UI data flow in a way no single contributor can. The LangSmith roadmap is its opposite: reflection-on-action - a structural admission that "it works on my machine" is not a defensible claim for an agent advising on healthcare costs. Designing the evaluation harness around quantifiable dimensions is my deliberate response to the stakeholder's challenge, even though it remains in development at the time of writing.

[Evaluate]

Two engineering principles have carried the integration phase. The first is ***Separation of Concerns***: because we agreed FastAPI endpoint contracts before any UI code was written, the frontend–backend connection has been smooth, and integration risk has concentrated where it warranted scrutiny - the agent's tool-routing and the calculator's branch logic. The second is ***dual-store discipline*****: SQLite for structured MBS, insurer, and clinic data, ChromaDB only for semantic policy retrieval,** with the agent always hydrating semantic hits back through SQLite. This isolation has kept technical debt local; a stale embedding cannot silently corrupt a rebate calculation. The limitations are equally honest. The NULL bug has revealed that our Definition of Done has been test-light on edge cases. A second limitation, surfaced by both the mentor and the tutor at the demo, is presentational rather than logical: **the agent's outputs are accurate but text-dense, leaving users unable to scan for the specific facts they need**. Most consequentially, the **LangSmith harness is *unfinished*;** until it produces quantified faithfulness and tool-selection metrics, the agent's reliability remains asserted rather than proven. Industry literature on production LLM systems (Huyen, Ng) is consistent that eval infrastructure is the most-underbuilt, highest-leverage component of any AI feature - a critique I now apply, with some discomfort, to my own deliverable.

[Plan]

The next moves over the closing weeks of the project are sharp. First, **complete the LangSmith evaluation harness** before the final external demonstration - targets are lowest possible hallucination rate, tool-selection accuracy above 90%, and low p95 latency - and pair it with an explicit edge-case checklist. Second, before final demo, **restructure the agent's output with headings, bulleted coverage breakdowns, and explicit key-value summaries** - addressing the mentor and tutor's shared feedback and **converting accurate retrieval into usable communication.** Longer-term, I intend to deepen my LLM-evaluation skill set with promptfoo, DeepEval, and LangSmith production-trace tooling, because the gap between *built it* and *can prove it works* is the gap between an MVP and an industry-grade product I want to close after this unit.

Reflection 3

AI Stewardship: Critical, Ethical Innovation

[Describe]

Across this semester, **AI has been a substrate, not a shortcut**, in every phase of MediBridge. I have used **Gemini Deep Research and NotebookLLM** to synthesise and document research on the MBS, OSHC, and Australian Privacy Principles regulatory environment, and **Claude for assisted coding of the AgenticRAG pipeline and the React frontend.** At **no point has AI been granted unsupervised authority**: the system design, knowledge-base structure, and database architecture were my decisions before any code was generated, and code generation itself has occurred under live supervision. The critical AI-stewardship incident came early in the project, during the research phase. My **initial queries returned outputs that either lacked citations entirely or summarised claims without surfacing the basis for them** - a structural hallucination risk in a domain where wrong information has medical consequences**.** I pivoted to deeper-reasoning workflows: Gemini Deep Research for traceable, multi-source synthesis, followed by manual verification against primary sources - the MBS XML, the OSHC Deed PDF, Department of Health fact sheets, and six insurer-specific knowledge files. At the architectural level, jointly with Abhay (our cyber-security lead) and with the tutor's subsequent confirmation, I proposed a stateless API with zero ingestion of policy numbers, names, contact details, or private health information. Two further safeguards now live inside the agent itself: every response carries a disclaimer instructing the user to verify with their specific OSHC provider, and the agent is constrained to explaining the Medicare–OSHC system, never offering medical advice. AI use has been disclosed to our industry mentor in conversation.

[Interpret]

Looking back across the project from this closing vantage point, the Gemini incident is more than a story about hallucination; it is a **meta-lesson about the difference between model *capability* and model *trustworthiness*.** A more fluent model that does not surface its reasoning is, in a healthcare-adjacent context, less trustworthy than a weaker model whose citations can be audited. **Trust is not earned by fluency; it is earned by auditability**. The stateless-API decision, which I proposed mid-project and which has shaped every architectural choice since, encodes the same principle structurally. Rather than mitigating privacy risk through policy ("we will handle data carefully"), we engineered the risk surface to *shrink*: **data we never collect cannot be breached, mis-disclosed, or misused** - and the easiest way to circumvent the obligations attached to policy numbers, names, contact data, and private health information is, simply, to never ingest them. This is *privacy by design* rather than privacy by promise, and it aligns directly with Australian Privacy Principle. The agent-level guardrails - verification disclaimers and a hard refusal to give medical advice - are AI safety guardrails in the strict alignment sense: the system's *refusal* behaviour is part of its specification, not an afterthought.

[Evaluate]

Evaluated from where I stand at the close of the project, the strengths of this stewardship posture are real and defensible. **Source-verification caught Gemini hallucinations before any unsupported claim entered the MediBridge knowledge base, and architectural data minimisation has removed an entire category of compliance and breach risk from the project.** The honest trade-off is that statelessness has cost MediBridge personalisation, cross-session memory, and the ability to learn from query patterns - a defensible cost for a regulated-domain MVP, but a real constraint future iterations will have to revisit thoughtfully, with explicit consent flows if Personal Health Information ever becomes necessary. A second honest limitation, visible to me only as I sit down to write this final reflection, is that AI disclosure to the industry mentor was not conveyed in detail or codified in writing; emerging professional norms - the EU AI Act's transparency obligations, NIST's AI Risk Management Framework, and Anthropic's own usage guidance - increasingly expect a *written* AI-use statement to accompany any AI-assisted deliverable. The triad these frameworks converge on is the same triad MediBridge already practised informally: *verification, minimisation, and disclosure*. I want the formal hygiene to match the informal discipline.

[Plan]

Three commitments anchor the next phase of my AI-stewardship practice, both for the closing tail of this submission and for the professional work that follows. First, I will publish a written AI-Use Statement with the final MediBridge deliverable, enumerating the tools used, the tasks delegated to each, the verification protocol applied, and the human-oversight points - converting the verbal disclosure already made to mentors into a permanent professional artefact before final submission. Second, I will adopt a *primary-source-or-nothing* rule for any AI-assisted research touching medical, legal, or financial content in future projects, and maintain a structured prompt-and-source log for full traceability. Third, I will elevate refusal behaviour to a first-class testable property of the agent - adding "out-of-scope queries and requests for medical advice are correctly declined" to the LangSmith evaluation suite I am currently building, so that ethical posture is measured, not assumed. The lesson MediBridge has taught me, and that I am taking out of this unit, is that AI literacy in 2026 is no longer about prompting well; it is about knowing when to disbelieve a confident model, and designing the system so that disbelief is structurally easy.

Appendix 1: Use of AI Declaration

Instructions

- **I did not use AI**. Complete only section A. Delete section B.
- **I used AI in an authorized way** (as described in the assignment brief). Complete only section B. Delete section A.
- **I am not sure**… please speak to your tutor before submitting the assignment.

Section A – Declaration that AI was not used:

I confirm that did not use AI at all for my assignment.

Section B – Evidence of AI use:

I confirm that I used AI in an authorized way (as described by the assignment brief).

Summary of AI use

(no word limit but we suggest 50-100 words describing how you used AI)

NA – No use of AI